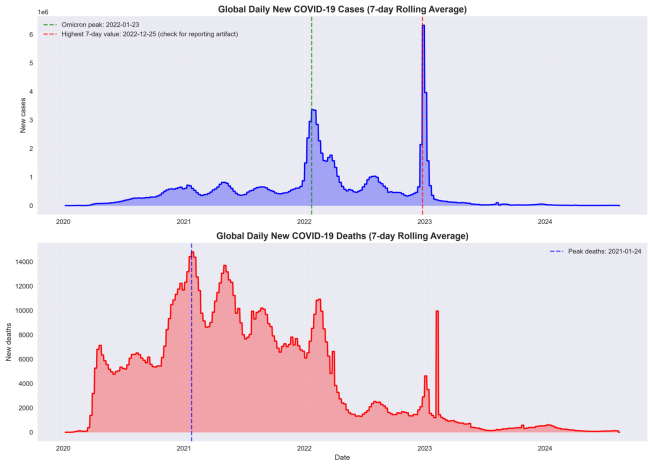
COVID-19 Global Impact Analysis - Executive Summary

Source: Our World in Data (to 14 Aug 2024) | 237 countries | Prepared by Archana, Data Analytics Intern | 15 Oct 2026

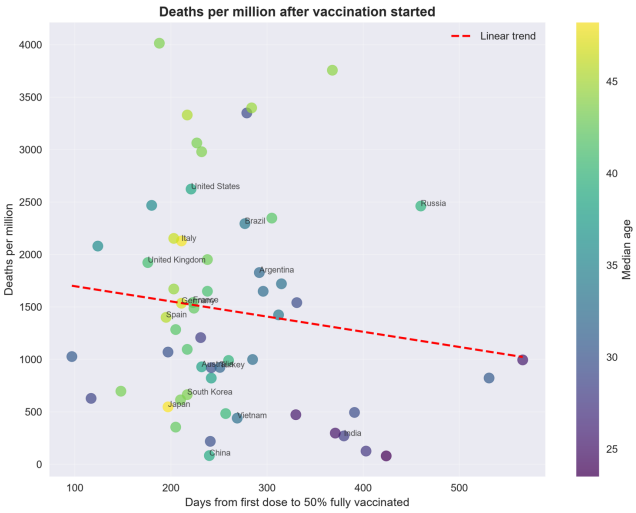
Question: did faster vaccination rollout go with lower death rates? Across 57 countries (over 1M people and 1M cases) we found **no statistically detectable relationship**: Spearman correlation between days to 50% fully vaccinated and deaths per million = -0.17 (p = 0.20); +0.07 after controlling for median age (p = 0.62). This does not show vaccines were ineffective: cumulative deaths depend heavily on age structure, reporting quality and timing, and country-level data cannot isolate vaccine effects.

Key findings

- **Deaths per reported case fell sharply.** Global deaths peaked at 14,811/day (2021-01-24); cases peaked at 3.36 million/day during Omicron (2022-01-23). Deaths per 1,000 reported cases fell from 24.5 to 2.5 between the two peaks. This is consistent with vaccination, prior infection and a milder variant, but testing and reporting also changed, so it is descriptive only.
- **Beware reporting artifacts.** The highest 7-day case value (6.3 million/day, Dec 2022) is not a global surge: China accounts for 86% of cases reported in that window, indicating a one-country reporting effect; China is also the largest contributor to the Feb 2023 death spike.
- **Case fatality rates vary widely.** Among 129 countries with over 100K cases, CFR ranges from 0.05% (Brunei) to 4.88% (Peru). Lowest: Brunei, Singapore, Iceland, South Korea, China. Highest: Peru, Egypt, Mexico, Bosnia and Herzegovina, Afghanistan. CFR reflects how many cases were detected, not only how deadly the disease was.
- **Rollout speed differed about eight-fold.** 87 countries (over 1M people) reached 50% fully vaccinated, from 97 days after the first dose (Israel) to 766 (Sierra Leone); median 257. Major countries: UK 176, Germany 211, US 221, France 223, Brazil 277, India 380 days.
- **Regional impact was uneven.** Asia reported 38.9% of cases and Europe 32.6%. Deaths per million: South America 3,104, Europe 2,815, North America 2,784, Oceania 731, Asia 347, Africa 182. Low figures in Asia and Africa likely partly reflect limited testing and reporting.
- **Early spread was fast.** The median doubling time of cumulative cases across countries was 9.6 days in March-April 2020.



Global daily cases and deaths (7-day average)



Rollout speed vs deaths per million after vaccination started (colour = median age)

Method and limitations

Method: removed aggregates and non-country rows; forward-filled cumulative columns within each country; derived 7-day averages, case fatality rate, vaccination rate, doubling time and days from first dose to 50% fully vaccinated; Spearman correlations with a median-age control. **Limitations:** underreporting and different testing strategies; one-country batch reporting creates spikes; rollout start is the first dose record in the data; country-level correlation is not causation.

Next steps

Compare deaths within age bands and matched time windows (for example, deaths per case before and after vaccination in the same country) and use the dataset's excess-mortality columns to reduce reporting bias.